

Java Streams

What is required for a Stream?

1. Data source

$\text{Stream} \langle T \rangle$  type of object to be processed

2. Chain of intermediate operations

intermediate operations are lazy



3. One terminal operation

terminal operations are eager

Data Sources: Examples

I get method

`java.util.Stream.Stream<T>.generate (Supplier<T>)`
`.of (T... items)`
• `Concat (Stream1, Stream2)`
• `empty ()`

`java.util.Collection: instance method .stream()`

`java.nio.file.Files.lines (path)`

Intermediate Operations: Examples

• `filter(Predicate <T>)` *one boolean method*

`1, 2, 3, 4` → `.filter((item) → item % 2 == 0)` → `2, 4`

• `map(Function <T, R>)`

`"CS400", "JAVA"` → `.map((item) → item.toLowerCase())` →

- `limit(int)`
- `skip(int)`

`"CS400", "java"`

Terminal Operations: Examples

- findFirst()
- forEach (Consumer<T>)  one void method
- count()
- min (Comparator<T>)
- max (Comparator<T>)
- reduce (T, BinaryOperator<T